

1.3 UK Environmental Challenges

The UK faces many challenges through people's interaction with the physical environment and the use of resources. This theme investigates some of the environmental challenges faced by the UK. Learners will look at extreme weather events in the UK, in particular

the links between extreme weather conditions and flooding. Learners will develop an understanding of the factors affecting the UK's energy use and security, the decision makers involved, as well as sustainability and management.

Section	Key Ideas	Content	Scale
1.3.1	The UK has a unique climate for its latitude which can create extreme weather conditions.	- How air masses, the North Atlantic Drift and continentality influence the weather in the UK. - How air masses cause extreme weather conditions in the UK, including extremes of wind, temperature and precipitation.	G, N
1.3.2	Extreme flood hazard events are becoming more commonplace in the UK.	Case study of one UK flood event caused by extreme weather conditions including: - causes of the flood event, including the extreme weather conditions which led to the event - effects of the flood event on people and the environment - the management of the flood event at a variety of scales.	N, R, L, F
1.3.3	Humans use, modify and change ecosystems and environments to obtain food, energy and water.	- Overview of how environments and ecosystems in the UK are used and modified by humans, including: - mechanisation of farming and commercial fishing to provide food - wind farms and fracking to provide energy - reservoirs and water transfer schemes to provide water.	N
1.3.4	There are a range of energy sources available to the UK.	- Identification of renewable and non-renewable energy sources. - The contribution of renewable and non-renewable sources to energy supply in the UK.	N, R, L
1.3.5	Energy in the UK is affected by a number of factors and requires careful management and consideration of future supplies.	- Changing patterns of energy supply and demand in the UK from 1950 to the present day, and how changes have been influenced by government decision making and international organisations. - Strategies for sustainable use and management of energy at local and UK national scales, including the success of these strategies. - The development of renewable energy in the UK and the impacts on people and the environment. - The extent to which non-renewable energy could and should contribute to the UK's future energy supply. - Economic, political and environmental factors affecting UK energy supply in the future.	G, N, R, L, F